RANJEET KUMAR

Master of Computer Applications
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Educational Qualification

Year	Degree/Examination	Institution/Board	CGPA/Percentage
2027	M.C.A	NIT, Trichy	07.61*
2024	B.C.A	T.P.S College , Patna ,Bihar	70.20%
2021	Class XII	M.L.K.S.S School Chaughara Supaul, BSEB	75.00%
2019	Class X	M.L.K.High School Chaughara Supaul, BSEB	83.00%

Internship Experience

- **Summer Internship – LEVELPRO Education LLP :** *May 2025 – Jun 2025*
  - Designed and developed a course-based learning platform for students.
  - Implemented workflows for student enrollment, structured content navigation, and course completion.
  - Gained hands-on experience in the EdTech product development lifecycle.
  - Worked on improving learner engagement and digital learning workflows.

Other Projects

- **FINMEM-X: AI-Powered Financial Decision Support Platform** *May 2026*
  - **Technologies:** Python, FastAPI, LangChain, LangGraph, Ollama, FAISS, PostgreSQL, TypeScript, Yahoo Finance API.
  - **Built** an AI-powered financial decision-support platform using **Multi-Agent Systems, RAG, and Hierarchical Memory** for personalized investment recommendations.
  - **Designed** six specialized agents for **News, Technical, Fundamental, Risk, Reflection, and Portfolio Analysis**, orchestrated through LangGraph shared state.
  - **Implemented** RAG with **FAISS** and hierarchical memory using **PostgreSQL + semantic retrieval** to leverage financial knowledge and previous investment experiences.
  - **Developed** a Decision Engine combining agent insights and portfolio analysis to generate **BUY/HOLD/SELL recommendations with confidence scores and explanations**

- **Digital Learning Platform:** *Mar 2026*

  - **Technologies:** React, TypeScript, Vite, Express.js, AWS DynamoDB, JWT, Google OAuth, Razorpay.
  - **Built** a full-stack EdTech platform with 17+ modules for courses, live lectures, tests, assessments, leaderboards, notes, chat, and admin management.
  - **Implemented role-based authentication** using JWT + Google OAuth, OTP email verification via Brevo API, and Razorpay payment integration.
  - **Developed** secure **RESTful APIs** with Express.js and DynamoDB, implementing RBAC, Helmet, CORS, and rate limiting.
  - **Engineered** performance tracking, student analytics, and scalable course-management workflows

- **DriveSafeAI: Multimodal Driver Drowsiness & Accident Risk Prediction System:** *Jan 2026*

  - **Technologies:** Python, YOLOv11, EfficientNet, LSTM, OpenCV, TensorFlow.
  - **Built** an AI-driven driver monitoring system using **Computer Vision, Deep Learning, and Sequential Learning** for real-time safety analysis.
  - **Implemented** video-based detection of **fatigue, distraction, eye closure, yawning, abnormal head movement, and phone usage**.
  - **Engineered** temporal risk assessment and inference pipelines using **YOLOv11, EfficientNet, and LSTM** for proactive accident-risk prediction.

Areas of Interest

- Data Structures and Algorithms
- Database Management System
- Object-oriented programming
- Operating System
- System Design

Technical Skills and Certifications

- Programming Languages : C, C++, Java Basics, Python, JavaScript
- Development : HTML, CSS, React, Node.js, Express.js, SQL, NoSQL
- Developer Tools : Visual Studio, Git, GitHub, Antigravity AI, Claude Code
- Artificial Intelligence : Machine Learning, Deep Learning, LLM, RAG

Positions of Responsibility

- **Member, Alumni Relationship Committee, Version'25:** *Mar 2025*
  A volunteer under the Version'25 (Member of the Alumni Relationship Committee), All India MCA meet which aims to bring the best students from M.C.A all over India to NIT, compete against each other and showcase their talent and potential.

Extracurricular Activities

- Represented C.A. department in **Football** during *Sportsfete 2024*, NIT Trichy
- Secured Third place in March Past in Sportsfete 2024 – the inter department sports fest organized by NIT Trichy.